

Hair biology and the hair-loss conversation you should have at 22

Week 4, Lecture 1, Looksmaxxing

Evidence-Based Looksmaxxing · Autumn 2026

What we will cover

- The hair growth cycle: anagen, catagen, telogen
- How androgenetic alopecia works and why it starts early
- Minoxidil: what the RCT evidence says (Olsen et al., 2002)
- Finasteride: what the systematic review says and what the side-effect data actually shows (Mella et al., 2010)
- When to see a dermatologist and what to ask

The hair growth cycle

Three phases, one follicle

- **Anagen (growth):** 2-7 years. Active cell division at the hair bulb. Scalp hair is in anagen 85-90% of the time.
- **Catagen (transition):** 2-3 weeks. Growth stops; the follicle shrinks.
- **Telogen (rest):** 3 months. The old hair sheds; a new anagen begins.

Each follicle runs its own clock independently of its neighbors.

Shedding 50-100 hairs per day is normal telogen cycling.

Follicular miniaturization: the AGA mechanism

- Dihydrotestosterone (DHT) is converted from testosterone by the enzyme 5-alpha reductase
- DHT binds to androgen receptors in susceptible follicles
- Each successive anagen phase grows a shorter, thinner, lighter hair
- Terminal hair miniaturizes to vellus hair, then to nothing
- The process is genetically programmed: follicle susceptibility is inherited, not caused by hats, stress, or shampoo

Androgenetic alopecia

Who gets AGA and when

- AGA affects approximately 50% of men by age 50 (Endotext, Trüeb, 2015)
- It can begin in the late teens or early twenties
- The Norwood-Hamilton scale classifies progression from Type I (no recession) to Type VII (horseshoe pattern)
- Vertex and frontal recession are the most common early presentations
- Finasteride works better at the vertex than at the frontal hairline (Endotext, Trüeb, 2015)

Why 22 is the right time to notice

- Miniaturization is easier to slow than to reverse
- FDA-approved treatments stabilize loss; they are not guaranteed to regrow dense hair after years of thinning
- A dermatologist can distinguish AGA from telogen effluvium (temporary shedding from stress, illness, or diet) and alopecia areata (autoimmune)
- Acting early, with evidence-based treatment, gives the best long-term outcome
- The goal is not panic; it is awareness

Minoxidil

The Olsen 2002 RCT: design and results

- Double-blind RCT, n = 393 men, 48 weeks
- Three arms: 5% topical minoxidil, 2% topical minoxidil, placebo
- Primary endpoint: mean change in non-vellus hair count per cm² at vertex

Group	Hair gain (non-vellus/cm ²)	Global improvement
5% minoxidil	+18.6	60%
2% minoxidil	+12.7	23% (placebo-matched)
Placebo	+3.9	23%

5% outperformed 2% by 45% and reached response four weeks earlier (Olsen et al., JAAD 2002).

How minoxidil works

- Minoxidil is a potassium channel opener: it dilates blood vessels in the scalp
- Mechanism on hair growth is not fully understood; current evidence points to prolonging anagen and increasing follicular blood flow
- Both topical (2%, 5%) and oral (low-dose) forms are used; oral minoxidil is not FDA-approved for AGA but is widely used off-label
- Dr. Sam Ellis (2022): oral minoxidil at low dose can be a convenient, affordable option when selected appropriately

What minoxidil does not do

- Does not block DHT: the root cause of AGA continues
- Results require ongoing use: stopping minoxidil returns the hair to its pre-treatment trajectory within months
- Shedding in the first 6-8 weeks is common as telogen hairs are displaced by new anagen growth: it is a sign the treatment is working, not causing more loss

Finasteride

The Mella 2010 systematic review: design

- Systematic review of 12 RCTs, n = 3,927 men
- Finasteride 1 mg/day vs placebo
- Published in Archives of Dermatology, October 2010
- Primary outcomes: hair count, investigator and patient global assessment
- Included assessment of adverse events: sexual side effects specifically

Mella 2010: efficacy results

- Mean hair count increase vs placebo:
 - Short-term: approximately 9.4%
 - Long-term: approximately 24.3%
- More finasteride patients rated as improved by investigators and patients both short- and long-term
- Evidence quality: moderate
- Finasteride works by inhibiting type-2 5-alpha reductase, reducing scalp DHT by approximately 60-70% (Endotext, Trüeb, 2015)

Mella 2010: side-effect data

- The review documents moderate-quality evidence of increased erectile dysfunction risk
- Sexual side effects including decreased libido and ejaculatory dysfunction were also reported
- Discontinuation rates due to side effects were comparable to placebo in the RCTs reviewed
- Post-finasteride syndrome (persistent side effects after stopping): the RCT data does not confirm or deny it at the population level; it is a real clinical report requiring honest acknowledgment
- The take: the sexual side-effect signal is real and documented; it is small but not zero; this is a decision to make with a prescribing dermatologist, not from a forum

Who should take finasteride

- Finasteride is a prescription medication in most countries
- See a dermatologist before starting: they will confirm the diagnosis, discuss your personal risk, and monitor for side effects
- Not appropriate for men with a family history of certain prostate conditions without medical review
- Combining minoxidil and finasteride is common practice; the Endotext chapter describes both agents as complementary rather than redundant

When to see a dermatologist

Red flags that warrant a visit

- Shedding that is sudden and dramatic (more than normal telogen cycling)
- Patchy hair loss rather than diffuse recession (possible alopecia areata)
- Hair loss alongside fatigue, weight change, or thyroid symptoms (possible telogen effluvium from systemic cause)
- Any interest in starting finasteride
- Uncertainty about whether what you are seeing is AGA or something else

What to say in the appointment

- "I think I may be experiencing androgenetic alopecia. Can you confirm the diagnosis and tell me what stage you are seeing?"
- "What treatment options would you recommend given my age and the stage of loss?"
- "If finasteride is appropriate, can you walk me through the side-effect risk so I can make an informed decision?"
- Bring your reference photos of your hairline over time if you have them

"The only US FDA-approved pharmacological treatments for androgenetic alopecia are topical minoxidil and oral finasteride. Finasteride is a type-2 5-alpha reductase inhibitor that decreases DHT conversion, with demonstrated miniaturization reversal and new hair growth."

<footer>Trüeb RM, Male Androgenetic Alopecia, Endotext/NCBI Bookshelf, 2015</footer>

Take-aways

1. AGA works by shortening the anagen phase through DHT-driven follicular miniaturization. It is genetic and can start in your early twenties.
2. 5% topical minoxidil produces 45% more non-vellus hair regrowth than 2% at 48 weeks (Olsen et al., 2002). Results require continuous use.
3. Finasteride increases mean hair count approximately 24% long-term versus placebo (Mella et al., 2010). The sexual side-effect signal is real, documented, and small.
4. Finasteride is a prescription decision to make with a dermatologist, not a self-prescribe from a forum.
5. Early awareness of AGA matters because stabilizing loss is easier than reversing advanced thinning.

What is next

- **Section (this week):** face-shape identification, reference photo selection, and barber brief
- **Lecture 2 (this week):** hair typing, wash frequency, face-shape-matched cuts, talking to your barber
- **Week 4 reading:** Hair, hair loss, and a haircut that fits your face
- **HW 2 out:** two-week skincare and grooming log